Smith et al. (2020)

Habitat Quality Assessment for Chinook Salmon

Interior British Columbia Streams

1. Spawning Substrate

Spawning substrate embeddedness is the dominant predictor of egg−to−fry survival for chinook salmon in interior BC streams. Field surveys at 47 index sites found that reaches where embeddedness exceeded 25% had egg−to−fry survival rates 62% lower than adjacent low−embeddedness sites. Fine sediment inputs from road crossings and livestock access were the primary drivers.

2. Riparian Shade and Stream Temperature

Riparian canopy removal raises maximum summer stream temperatures by 3 to 7 degrees Celsius in small to medium channels. In reaches where streamside conifers were harvested within 30 m of the bankfull channel, daily maximum temperatures increased by an average of 4.2 degrees C compared to unlogged reference reaches. Temperature elevations of this magnitude exceed the upper thermal tolerance of juvenile chinook during late−summer rearing.

3. Road Crossings and Fish Passage

Fish passage assessments at 1,200 road−stream crossings found that 38 percent were rated as full or partial barriers to adult chinook migration. Culverts with outlet drops exceeding 15 cm or insufficient water depth during low flow accounted for 71% of all barrier crossings. Barrier removal restored upstream colonisation within 1 to 3 years of installation.

4. Large Woody Debris

Pool density and large woody debris loading were positively correlated across all 47 assessment reaches. Sites with more than 25 pieces of wood per 100 m of channel had 2.8 times more pool−associated fish than reaches with fewer than 5 pieces per 100 m. Wood recruitment from adjacent riparian conifers was the primary driver of debris loading.